11/2/26 attendance →



AI Embedded HIL

UNIVERSITY of WASHINGTON

Contributing to the [GitHub repository]

- **Accept the invite:** check your email or GitHub notifications (you can't push code until you accept the invite)
- **Clone the repo in a VSCode terminal:**
  git clone https://github.com/TBC-Projects/AI-Embedded-HIL.git
- **Create your own branch:**
  git checkout -b <yourname/feature-name>
- **Commit and push:**
  git add .
  git commit -m "<Short description of change>"
  git push -u origin <yourname/feature-name>
- **Open a pull request** on GitHub: ***base = main, compare = your branch***
- **Request a review** from a lead: ***@Yd025 @eba14 @RuiqiLiu2014***
- **A lead reviews and merges** into main. Never push directly to main

Getting a 403 error? Make sure you accepted the invite, then run **gh auth login** in your VSCode terminal to sign in with the right account.

Software Team

Last week:

- Discussed the project plan and team roles
- Started on the frontend/UI of the dashboard

This week:

- **Repo access:** accept the GitHub invite, clone, and work from your own branch
- **Architecture walkthrough:** how the pieces fit together (prompt → generate → compile → QEMU boot check) and which folder each part lives in
- **Tasks:**
  - **Agent loop**: feed compile/QEMU errors back to the agent and retry
  - **Backend**: job queue endpoints (submit job, check status)
  - **Dashboard**: build on last week's frontend to submit a prompt and watch progress
  - **Runner**: send a compiled binary to the Pi over UART
- **Interface check-in with hardware:** agree on the JSON format for FPGA results
- **Hands-on:** blink the Pi's green LED (next slides)



Software Team Deliverable Resources

- [Raspberry Pi Blinking Instructions] (Done)
- [Orchestration]
- [Flashing Protocol]
- [FPGA Results Format] (Depends on Hardware Team's FPGA work)

Hardware Team

- Deploy something on board
- Sending FPGA data packets to computer
- Reading Raspberry Pi data and wiring FPGA to Pi
- More board stuff for later: https://github.com/Digilent/vivado-boards/tree/master

Short Vivado guide

1. Add sources (copy to project)
2. Add constraints (copy to project)
3. Run synthesis
4. Run implementation
5. Generate bitstream
6. Open hardware manager
7. Open target
8. Program device
9. Celebrate



References

- *Project inspired by: [Sentinel] (read this!)*

- FPGA Board: [Amazon]
- Raspberry Pis: [Tinkersphere]
- USB-to-UART adapter: [Shillehtek]
- MicroSD Card: [Amazon]

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Thanks for coming!

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